

Simple Linear Regression – Part-1

Objective

To understand and implement Simple Linear Regression using Python and Machine Learning.

Mathematical Model

$$y = mx + c$$

- **x**: Independent variable
- **y**: Dependent variable
- **m**: Slope
- **c**: Intercept

```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

Step 1: Create Dataset

We create a small dataset representing study hours and exam scores.

```
# Create a simple dataset containing study hours and corresponding scores
data = {
    "Hours": [1, 2, 3, 4, 5, 6, 7, 8], # Independent variable: number of hours studied
    "Score": [35, 40, 50, 55, 65, 70, 75, 85] # Dependent variable: score obtained
}
```

```
# Convert the dictionary into a Pandas DataFrame for easy analysis and modeling
df = pd.DataFrame(data)
```

```
# Display the dataset- first 5 rows
df.head()
```

	Hours	Score
0	1	35
1	2	40
2	3	50
3	4	55
4	5	65

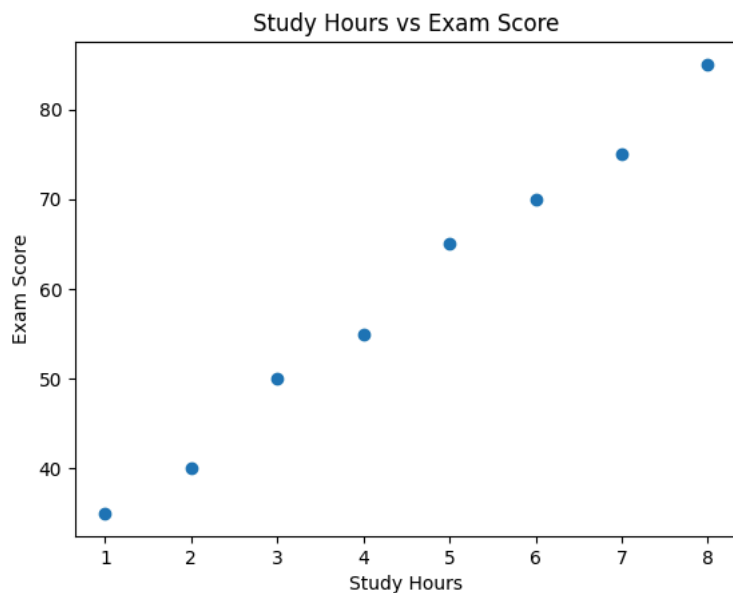
Next steps: [Generate code with df](#) [New interactive sheet](#)

Step 2: Define Independent and Dependent Variables

```
# Independent and dependent variables
X = df[["Hours"]]
y = df["Score"]
```

Step 3: Visualize the Dataset

```
plt.scatter(df['Hours'], df['Score'])
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.title('Study Hours vs Exam Score')
plt.show()
```



Step 4: Train the Linear Regression Model

```
model = LinearRegression()
model.fit(X, y)
```

LinearRegression ⓘ ?

Simple Linear Regression on CSV Dataset

Objective

To implement Simple Linear Regression to predict Placement Package (LPA) based on CGPA.

This is formatted as code

Part 1: Train and Save the Model

Step 1: Import Required Libraries

```
# Import pandas for data handling
import pandas as pd

# Import matplotlib for data visualization
import matplotlib.pyplot as plt

# Import Linear Regression model
from sklearn.linear_model import LinearRegression
```

Step 2: Load Dataset from CSV File

```
# Read the dataset from CSV file
df = pd.read_csv('/content/drive/MyDrive/Dataset/Placement.csv')
```

Step 3: Understand the Dataset

```
# Display the shape of the dataset (rows, columns)
df.shape
```

```
(8, 2)
```

```
# Display dataset information such as column names and data types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Hours   8 non-null        int64
 1   Score   8 non-null        int64
dtypes: int64(2)
memory usage: 260.0 bytes
```

```
# Display statistical summary of the dataset
df.describe()
```

	Hours	Score
count	8.00000	8.000000
mean	4.50000	59.375000
std	2.44949	17.410485
min	1.00000	35.000000
25%	2.75000	47.500000
50%	4.50000	60.000000
75%	6.25000	71.250000
max	8.00000	85.000000

```
df.head()
```

	cgpa	package
0	6.89	3.26
1	5.12	1.98
2	7.82	3.25
3	7.42	3.67
4	6.94	3.57

Step 5: Define Independent and Dependent Variables

```
# Independent variable (input feature)
X = df[['cgpa']]

# Dependent variable (target)
y = df['package']
```

Step 6: Split Data into Training and Testing Sets

```
# Import train_test_split function
from sklearn.model_selection import train_test_split

# Split the data (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=2
)
```

Step 7: Create and Train Linear Regression Model

```
# Create Linear Regression model object
model = LinearRegression()

# Train the model using training data
model.fit(X_train, y_train)
```

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LinearRegression()

Prediction Using User Input

```
# Take CGPA value from the user
cgpa = float(input("Enter CGPA: "))

# Convert input into 2D array format required by the model
cgpa_input = [[cgpa]]

# Predict the placement package using the trained model
predicted_package = model.predict(cgpa_input)

# Display the prediction
print("Entered CGPA:", cgpa)
print("Predicted Package (in LPA):", predicted_package[0])
```

```
Enter CGPA: 5
Entered CGPA: 5.0
Predicted Package (in LPA): 1.8936479448824457
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
warnings.warn(
```

Save the Model

```
import pickle
```

```
pickle.dump(model, open('SLRModel.pkl', 'wb'))
```

Part 2: Create Flask Backend

```
from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

model = pickle.load(open("SLRModel.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():
    cgpa = float(request.form['cgpa'])
    prediction = model.predict(np.array([[cgpa]]))

    return render_template(
        "index.html",
        prediction_text=f"Predicted Package : {prediction[0]:.2f} LPA"
    )

if __name__ == "__main__":
    app.run(debug=True)
```

Part 3: Create the HTML Frontend

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Package Predictor</title>

    <style>

        *{
            margin:0;
```

```
padding:0;
box-sizing:border-box;
font-family:Arial, Helvetica, sans-serif;
}

body{
background:#f4f7fc;
display:flex;
justify-content:center;
align-items:center;
height:100vh;
}

.container{

width:420px;
background:white;
padding:35px;
border-radius:15px;
box-shadow:0px 8px 20px rgba(0,0,0,0.2);
text-align:center;

}

h1{

color:#0d47a1;
margin-bottom:10px;

}

h3{

color:#555;
margin-bottom:25px;

}

label{

display:block;
text-align:left;
font-weight:bold;
margin-bottom:8px;

}

input{

width:100%;
padding:12px;
font-size:16px;
border:1px solid #ccc;
border-radius:8px;
margin-bottom:20px;

}

button{

width:100%;
padding:12px;
background:#1565c0;
color:white;
border:none;
border-radius:8px;
font-size:18px;
cursor:pointer;
transition:0.3s;

}

button:hover{

background:#0d47a1;

}

.result{

margin-top:25px;
padding:15px;
```

```
        border-radius:8px;
        background:#e8f5e9;
        color:green;
        font-size:20px;
        font-weight:bold;

    }

    footer{

        margin-top:20px;
        font-size:13px;
        color:#777;

    }

</style>

</head>

<body>

<div class="container">

    <h1>Student Package Predictor</h1>

    <h3>Simple Linear Regression</h3>

    <form action="/predict" method="POST">

        <label>Enter Student CGPA</label>

        <input
            type="number"
            name="cgpa"
            step="0.01"
            min="0"
            max="10"
            placeholder="Example : 8.45"
            required>

        <button type="submit">
            Predict Package
        </button>

    </form>

    {% if prediction_text %}
    <div class="result">
        {{ prediction_text }}
    </div>
    {% endif %}

    <footer>

        Machine Learning by Mr. Aditya Tanaji Gawade

    </footer>

</div>

</body>
```